Multimedia Signal processing

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1. **Question (8 pts)**:

   Consider a sequence of binary simbols *aabaaabaaaaabbaaaaba* emitted by a source X.

   (a) Learn the p.m.f. of the source from the observed data and compute the entropy of the source (assuming a memoryless source).

   (b) Learn the conditional p.m.f. $Pr(X_n|X_{n-1})$.

   (c) Estimate the number of bits needed to encode the sequence *aaaba* using Elias coding in two cases: i) without exploiting source memory; ii) exploiting source memory.

   Solution

   (a) The p.m.f. can be estimated as follows

   $$Pr(X = a) = 15/20 = 3/4 \tag{1}$$
   $$Pr(X = b) = 5/20 = 1/4 \tag{2}$$

   The entropy of the source is

   $$H(X) = -3/4 \log_2 3/4 - 1/4 \log_2 1/4 \tag{3}$$

   (b) The conditional probabilities can be calculated as follows

   $$Pr(X_n = a|X_{n-1} = a) = 10/14 \tag{4}$$
   $$Pr(X_n = b|X_{n-1} = a) = 4/14 \tag{5}$$

   $$Pr(X_n = a|X_{n-1} = b) = 4/5 \tag{6}$$
   $$Pr(X_n = b|X_{n-1} = b) = 1/5 \tag{7}$$

   (c) The number of bits necessary to perform Elias coding is $\lceil \log_2 p(\mathbf{x}_N) \rceil + 1$.
   In the first case,
   $$Pr(\mathbf{x}_N) = Pr(a)^4 \cdot Pr(b) = (3/4)^4 \cdot (1/4) = 0.0791 \tag{8}$$
   Hence, $L = 5$ bits. In the second case,

   $$Pr(\mathbf{x}_N) = Pr(a) \cdot Pr(a|a) \cdot Pr(a|a) \cdot Pr(b|a) \cdot Pr(a|b) = 3/4 \cdot 10/14 \cdot 10/14 \cdot 4/14 \cdot 4/5 = 0.0875 \tag{9}$$

   Hence, $L = 5$ bits.

2. **Question (5 pts)**

(a) Illustrate the architecture of motion-compensated predictive video coding.

(b) Discuss the need for a closed-loop architecture.

Solution

See slide set `MMSP_3.2_Video coding`.

3. **Question (5 pts)** Illustrate the KLT transform, its design principles, optimality properties and drawbacks.

   Solution

   See slide set `MMSP_1.4_Transform_Coding`.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider an audio coding system

   - [T] [F] Perceptual audio coding can be used to encode high-quality speech.
     TRUE. Perceptual audio coding does not assume a priori a model for the source to be encoded. Hence, it can also be applied to speech signals.
   - [T] [F] Quantization is applied directly in the time domain.
     FALSE. Audio coding adopts transform coding. As such, quantization is applied in the transform domain (e.g., after PQMF, MDCT)
   - [T] [F] The MDCT achieves perfect reconstruction when the length of the analysis window is kept constant.
     FALSE. The analysis window can be varied in length, provided that the TDAC conditions are satisfied at the boundary between consecutive windows.
   - [T] [F] The perceptual model is encoder-dependent.
     TRUE. The perceptual model is not defined by the standard. Hence, each encoder implementation can follow its own guidelines in devising the perceptual model.

   (b) Consider a discrete source

   - [T] [F] $L_S(x) \geq L_H(x)$, where $L_S(x)$ and $L_H(x)$ denote, respectively, the average code length of Shannon and Huffman coding for a discrete memoriless source x
     TRUE. Huffman coding is optimal. Hence, Shannon coding can be as good as Huffman coding.
   - [T] [F] $L_S(x) = L_H(x)$, if and only if the p.m.f. is dyadic.
     TRUE. Shannon coding is optimal for dyadic p.m.f.'s, since $L_S = H(x)$.
   - [T] [F] The code length of each codeword is upper bounded by $H(x) + 1$
     FALSE. The average code length is upper bounded by $H(x) + 1$. Some codewords can be longer than that.

   (c) Consider a quantization scheme based on compander-expander.

   - [T] [F] The quantization symbols are equally likely.
     TRUE. The compander maps the p.d.f. of the source to a uniform p.d.f., which is uniformly quantized.
   - [T] [F] The mean square error distortion is less than that achievable by the Lloyd-Max quantizer, for the same number of reconstruction levels.
     FALSE. The L-M quantizer minimizes the MSE.
   - [T] [F] The input source x is mapped to an intermediate source y by means of the mapping $y = x^{\alpha}$, $\alpha \in \mathbb{R}$
     FALSE. The mapping is equal to the c.d.f. of the source.

5. **Question (Matlab - 10pts)**

   Image tampering detection and localization can be performed exploiting the traces left by JPEG compression.

   A forensic analyst wants to analize image *image.png*, which is available in a lossless compression format.

   (a) Determine whether the image was previously compressed with JPEG. To this end, perform the block-wise 2D-DCT transform, with blocks of size 8×8, and analyze the histograms of DCT coefficients. Comment your results.

   (b) Determine (by visual inspection) the quantization step sizes used for the DCT coefficients $(2, 3)$ and $(4, 4)$.

(c) Shift the image by 1 pixel both horizontally and vertically. Repeat the previous steps on the shifted image. Comment your results.

(d) As an alternative method, compress the image *image.png* using different JPEG quality factors QF (Hint: use `imwrite(Image, filename, 'Quality', q)`). Compute the PSNR between the image *image.png* and the compressed images. Plot the curve PSNR vs QF and comment your results.

Solution

```
%% 2013_07_03_MMSP_exam

clear all
close all
clc

%% Part (a), (b) and (c)
PART = 'partA';

switch PART
    case 'partA'
        Image = double(imread('image.png'));
        [Nr, Nc] = size(Image);
    case 'partC'
        Image = double(imread('image.png'));
        [Nr, Nc] = size(Image);
        ImagePadded = padarray(Image, [1 1], 'replicate', 'pre');
        Image = ImagePadded(1:Nr, 1:Nc);
end



N = 8;

Image_dct = zeros(N,N,Nr*Nc/N^2);

% for each block size
k = 1;
for br = 1:Nr/N
    for bc = 1:Nc/N
        % consider a block of size n x n
        block = Image((br-1)*N+1 : br*N, (bc-1)*N+1 : bc*N);
        % compute the DCT
        block_dct = dct2(block);
        Image_dct(:,:,k) = block_dct;
        k = k+1;
    end
end

figure(1)
subplot(121)
temp = Image_dct(3,2,:);
hist(temp(:), 10000)
subplot(122)
temp = Image_dct(4,4,:);
hist(temp(:), 10000)

% (a)
% Histogram peaks are distinctly visible at equispace positions. This is a
% clear sign of scalar uniform quantization due to JPEG compression

% (b)
% (3,2): histogram peaks are spaced Qstep = 5 apart
% (4,4): histogram peaks are spaced Qstep = 12 apart
% Larger quantization step sizes are used at higher frequencies.

% (c)
% By shifting the image, the histogram does not show the characteristic
% comb-shaped pattern, although the image was compressed with JPEG. This is
% due to the mis-alignment between the original JPEG blocks and the grid
% used by the analyst.


%% Part (d)
for i = 1:100
    imwrite(Image / 255, 'temp.jpg', 'Quality', i);
    ImageRecompressed = double(imread('temp.jpg'));

    PSNR(i) = 10*log10(255^2 ./ (sum(sum((Image - ImageRecompressed).^2))/(Nr*Nc)));

end

figure(2)
plot([1:100], PSNR);
% (d) A sharp peak is clearly visible at QF = 80. This is an indication
% that the image was originally compressed at that quality factor.
```